

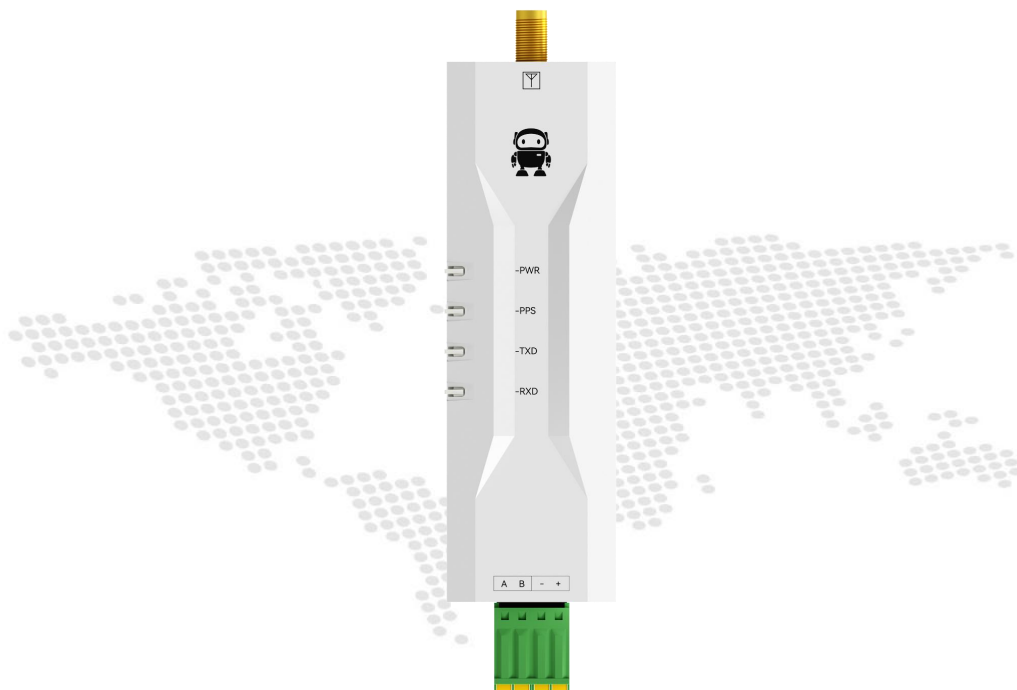


Chengdu Ebyte Electronic Technology Co.,Ltd

Wireless Modem

User Manual

EWD108-GN0x



All rights to interpret and modify this manual belong to
Chengdu Ebyte Electronic Technology Co., Ltd.

Contents

Disclaimer and Copyright Notice.....	3
1 Introduction.....	4
1.1 Brief Introduction.....	4
1.2 Features.....	4
1.3 Application.....	4
2 Technical indicators.....	5
2.1 General specification parameters.....	5
2.2 GNSS Performance.....	6
2.3 Mechanical dimension drawings.....	7
2.4 Pin and Indicator Definitions.....	7
2.5 Installation method.....	8
3 Modbus Registers.....	9
3.1 Communication protocols.....	9
3.2 Register Description.....	9
3.3 Register Table.....	9
4 Quick Start.....	12
4.1 Preparation work.....	12
4.2 Demonstration of use.....	12
5 Modbus RTU Usage.....	14
5.1 Holding registers.....	14
5.2 Read Holding Register Instruction.....	15
5.2.1 Read reserved bit 0001.....	15
5.2.2 Read device address (broadcast).....	15
5.2.3 Reading device baud rate.....	15
5.2.4 Parity Reading.....	16
5.2.5 Reading positioning data (RMC).....	16
5.2.6 Reading antenna settings.....	17
5.2.7 Reading the product model.....	18
5.2.8 Reading version information.....	18
5.2.9 Reading the Satellite System.....	18

5.2.10 Read the update frequency of GNSS positioning.....	19
5.3 Write Holding Register Instructions.....	19
5.3.1 Modify device address (broadcast).....	19
5.3.2 Modifying the baud rate.....	19
5.3.3 Modifying the parity bit.....	20
5.3.4 Modifying antenna settings.....	20
5.3.5 Modifying the IAP flag.....	20
5.3.6 Modifying the Satellite System.....	21
5.3.7 Adjust the update frequency of the GNSS module.....	21
Revision history.....	21
About us.....	22

Disclaimer and Copyright Notice

The information in this document, including the URL addresses for reference, is subject to change without notice. this document is provided ‘as is’ without warranty of any kind, including any warranty of merchantability, fitness for a particular purpose, or non-infringement, and any warranty that any proposal, specification, or sample is referenced elsewhere. This document disclaims all liability, including liability for infringement of any patent rights, arising out of the use of the information in this document. No licence, express or implied, is hereby granted by way of estoppel or otherwise, to use any of the intellectual property rights in this document.

The test data in this document are obtained from EBYTE’s laboratory tests and the actual results may vary slightly.

All trade names, trademarks and registered trademarks mentioned in this document are the property of their respective owners and are hereby declared.

The final interpretation right belongs to Chengdu Ebyte Electronic Technology Co.

Note: The contents of this manual are subject to change due to product version upgrade or other reasons. EBYTE reserves the right to make changes to the contents of this manual without any notice or prompting. EBYTE makes every effort to provide accurate information in this manual, however, EBYTE does not warrant that the contents of this manual are error-free, and all statements, information and recommendations in this manual do not constitute any warranty, express or implied.

1 Introduction

1.1 Brief Introduction

EWD108-GN0x series is a positioning terminal that supports multiple positioning standards (GPS, BDS, GLONASS, Galileo, etc.) with fast response and accurate positioning.

The equipment outputs positioning information via the Modbus RTU protocol. It supports outputting RMC-ASCII strings in accordance with the NMEA0183 protocol and separately stores individual registers (longitude, longitude direction, latitude, latitude direction, etc.). The maximum serial port baud rate can reach 115200 bps. It can be easily modified via the Modbus RTU protocol and supports RS485/RS232/TTL communication interfaces, which is convenient and fast.



1.2 Features

- Supports BDS/GPS/GLONASS/GALILEO/QZSS/SBAS satellite navigation systems, as well as multi-system combined positioning (different models support different positioning systems);
- EWD108-GN03 series supports satellite systems: BDS/GPS/GLONASS, with positioning accuracy up to 2.5 meters (CEP50);
- EWD108-GN03B series supports satellite systems: BDS, with positioning accuracy up to 2.5 meters (1 σ);
- EWD108-GN04 series supports satellite systems: BDS/GPS/GLONASS/GALILEO, with positioning accuracy up to 1.5 meters (CEP50);
- EWD108-GN05 series supports satellite systems: BDS/GPS/GLONASS/GALILEO/QZSS/SBAS, with positioning accuracy up to 1.0 meter (CEP50);
- EWD108-GN06B series supports satellite systems: BDS, with positioning accuracy up to 1.0 meter (CEP50);
- Various serial port baud rates (1200-115200bps) can be configured;
- Support standard Modbus RTU protocol to read positioning information;
- Multiple positioning information output formats;
- Output antenna positioning status through registers and indicators;
- Support active antenna and passive antenna switching;
- The communication interface is equipped with TVS protection;
- Support wide voltage input DC5~24V;
- Supports rail mounting and positioning hole installation;
- Industrial-grade design, support operating temperature up to -40 ~ 85°C;

1.3 Application

- In-vehicle positioning and navigation equipment;
- Industry equipment with a need for GNSS positioning or navigation;
- Asset tracking, etc;

2 Technical indicators

2.1 General specification parameters

General Specification Parameters					
Item	EWD108-GN03 (xxx)	EWD108-GN03B (xxx)	EWD108-GN04(xxx)	EWD108-GN05(xx x)	EWD108-G N06B(xxx)
Supply Voltage	5V~24V DC				
Communications interface ¹	RS485/RS232/UART				
Baud rate range	1200-115200bps				
Communication protocols	Modbus RTU				
Supported positioning systems	BDS/GPS/GLO NASS	BDS	BDS/GPS/GLON ASS/GALILEO	BDS/GPS/GLONA SS/GALILEO/QZS S/SBAS	BDS
Operating frequency	BDS:B1 GPS:L1 GLONASS:L1	BDS:B1C, B1I	GPS:L1 C/A QZSS:L1 C/A/S GLONASS:L10F BDS:B1I, B1C GALILEO:E1B/C SBAS:L1 C/A(WAAS,EGN OS,MSAS,GAGA N)	GPS/QZSS:L1 C/A, L1C,L2C,L5,L6 BDS:B1C,B1I,B2a, B2I,B3I GLONASS:L1,L2 Galileo:E1,E5,E6 SBAS:WAAS,EGN OS,MSAS,GAGAN ,SDCM	BDS:B1I,B1 C,B2I,B3I,B2 a,B2b
Positioning update rate	1Hz~10Hz				
User configuration	Modify registers via Modbus RTU, reboot to take effect				
Antenna Interface	SMA (Internal hole for external thread)				
Hardware interface	4Pin 3.81mm Phoenix Terminal				
Size	103×28×28.5mm (L×W×H)				
weight	32±5g				
Operating temperature	-40 ~ +85°C, Industrial grade				
Operating humidity	10% ~ 90%, Relative humidity, non-condensing				

¹ EWD108-GN0x (485) communication interface: RS485; EWD108-GN0x (232) communication interface: RS232;
EWD108-GN0x (TTL) communication interface: UART.

Storage temperature	-55°C-100°C
---------------------	-------------

2.2 GNSS Performance

GNSS Performance Parameter Table							
Type	Indicator term	EWD108-GN03(xxx)	EWD108-GN03B(xxx)	EWD108-GN04(xxx)	EWD108-GN05(xxx)	EWD108-GN06B(xxx)	Unit
Positioning time (test condition 1)	Cold-start	32	23	28	27.5	23	s
	Warm-start	1	1	1	1	1	s
	Recapture	1	1	1	1	1	s
Sensitivity (test condition 2)	Cold-start	-148	-148	-148	-148	-148	dBm
	Warm-start	-156	-156	-159	-162	-156	dBm
	Recapture	-160	-160	-160	-164	-160	dBm
	Trace	-162	-162	-167	-166	-162	dBm
Accuracy (test condition 3)	Horizontal positioning accuracy	2.5	2.5(level)/3.0(vertical)	1.5	1.0	1.0	m
	Speed Positioning Accuracy	0.1	0.1	0.05	0.1	0.05	m/s
	Timing accuracy	30	30	30(RMS)/60(99%)	30	10	ns
Power Consumption (test condition 4)	Capture Current	15	20	10	20	20	mA
	Tracking current	15	20	10	20	20	mA

[Note]: The above results refer to the GPS/Beidou dual-mode operation mode or the single Beidou operation mode.

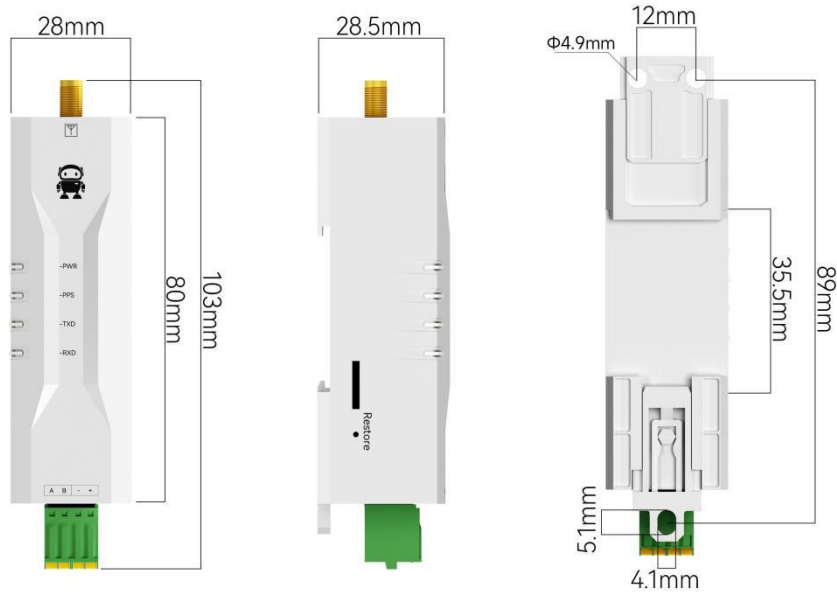
[Test Condition 1]: The number of received satellites is more than 6, all satellite signal strength is -130dBm, 10 times testing to take the average value, positioning error less than 10 metres.

[Test condition 2]: External LNA noise factor 0.8, the number of receiving satellites is more than 6, the received signal strength value under the condition of locking or non-locking within five minutes.

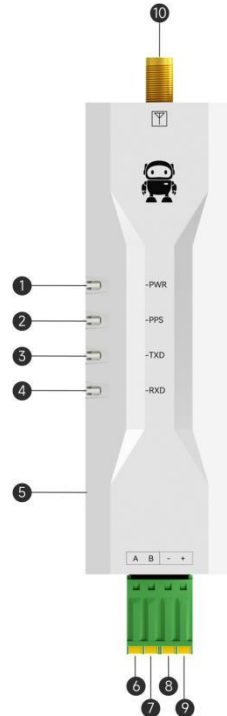
[Test Condition 3]: Open and unobstructed environment, 24 hours continuous power-on test, 50% CEP.

[Test Condition 4]: Power supply voltage DC 12V. The number of receiving satellites is more than 6. The signal strength of all satellites is -130dBm.

2.3 Mechanical dimension drawings



2.4 Pin and Indicator Definitions

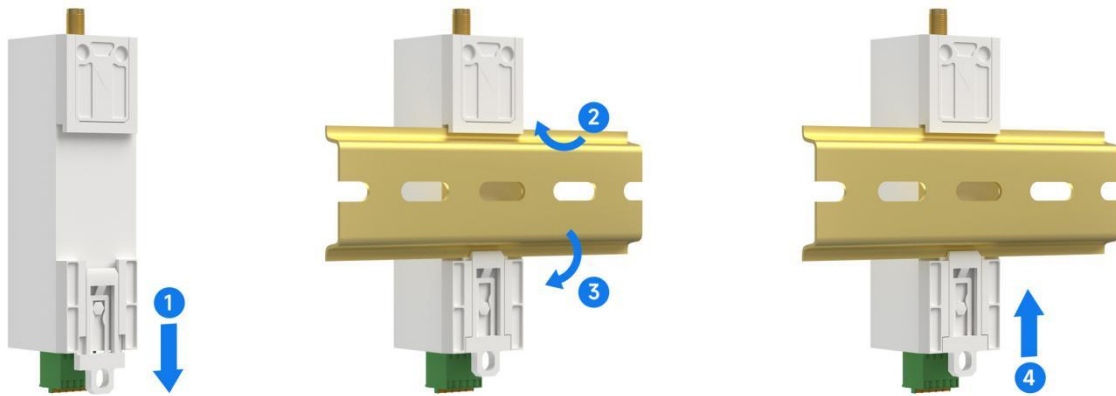


Serial Number	Name	Functional Description
1	PWR indicator light	Power indicator, power-on status light is always on

2	PPS Seconds pulse indicator	When positioning is invalid, it remains off; when positioning is valid, it flashes once per second.
3	TXD indicator light	Transmit indicator, blinks when sending data to the RS485 bus
4	RXD indicator light	Receive indicator, blinks when receiving data from the RS485 bus
5	Factory Key	Long press 5-10s effectively to restore the device address and baud rate parameters to factory parameters; Factory parameters: device address is 1, serial port parameters 9600/8/no parity/1
6	A/TXD	RS485 bus's A interface/UART_TXD/RS232 interface TXD
7	B/RXD	RS485 bus's B interface/UART_RXD/RS232 interface RXD
8	Power Negative	POWER GROUND
9	Power Positive	Power Positive Connector, 5~24V DC
10	ANT	SMA antenna connector

2.5 Installation method

The equipment is mounted on rails and positioning holes.



Rail mounted

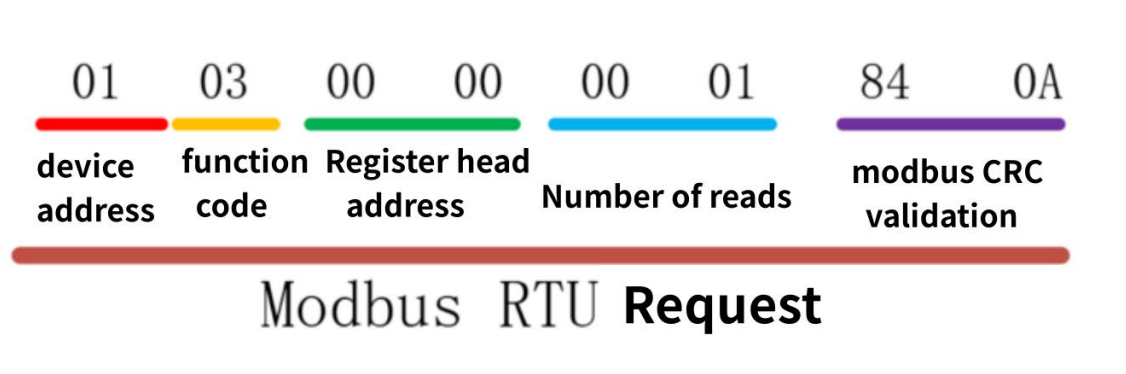
3 Modbus Registers

3.1 Communication protocols

The protocol layer of GPS/BeiDou positioning module is the standard Modbus communication protocol, which is in line with the national standard GBT 19582.1-2008 “Specification for Industrial Automation Network Based on Modbus Protocol”. It adopts Modbus RTU communication protocol, which receives and parses the frame data on the data bus, and

returns the data according to the results of the parsing.

The frame format is as follows:



3.2 Register Description

The following table describes the specific meaning of the decimal registers and the function codes used for their operation.

Register	Read function code (HEX)	Write function code (HEX)	Examples of channels
[Area 1] Discrete Input Register	02	—	10001, Indicates the DI1 register address
[Area 0] Switching Output Register	01	05 0F	00001, Indicates the DO1 register address
[Area 3] Input register	04	—	30001, Indicates zone 3 address 1
[Area 4] Holding register	03	06 10	40001, Indicates Zone 4 address 1

Note: This device will only use Area 4 (i.e., the Holding Register) register addresses.

3.3 Register Table

Register function	Register address		Data format	Data range/remarks
	(Decimal)	(Hexadecimal)		
Reservation (RES)	40001	0001	Int16	None, invalid data readout
Device address	40002	0002	Int16	1-255, default 1, support broadcast address (0x00) reading and modification
Baud rate	40003	0003	Int16	波特率代码: 0x0000: 1200bps, 0x0001: 2400bps, 0x0002: 4800bps, 0x0003: 9600bps, 0x0004: 19200bps, 0x0005: 38400bps, 0x0006: 57600bps, 0x0007: 115200bps,
Verification Methods	40004	0004	Int16	0x0000: No parity, 0x0001: Odd Parity, 0x0002: Even parity.

				Configuring other parameters does not take effect
RMC-Positioning data	40005	0005	String (70Byte)	Stores 70 bytes of RMC- NMEA0183 protocol data, encoded in ASCII and decoded in AB order
...	...	...	...	reservation
Position state	40200	00C8	Int16	0: Positioning not valid, 1: Positioning valid, read-only
Year	40201	00C9	Int16	2022 stands for the year 2022, read only
Month	40202	00CA	Int16	Values range from 1 to 12, representing January to December, respectively, and are read-only
Day	40203	00CB	Int16	Values range from 1 to 31, representing numbers 1 to 31, respectively, and are read-only
Hour	40204	00CC	Int16	The values range from 0 to 23, representing 0:00 to 23:00, respectively, and are read-only
Minute	40205	00CD	Int16	The values range from 0 to 59, representing 0 to 59 points, respectively, and are read-only
Seconds	40206	00CE	Int16	Values range from 0 to 59, representing 0 to 59 seconds, respectively, read-only
Longitude direction	40207	00CF	Int16	0x45 (ASCII: E) for east longitude. 0x57 (ASCII: W) for west longitude, read only
Longitude	40208	00D0	Float (4Byte)	The unit is degree, 5 decimal places, read-only. Example: 103.93416°, word order: big endian, byte order: big endian
Latitudinal direction	40210	00D2	Int16	0x4E (ASCII: N) represents north latitude, read-only 0x53 (ASCII: S) for South Latitude
Latitudinal	40211	00D3	Float (4Byte)	Units in degrees, 5 decimal places, read-only Example: 30.77056, word order: big endian, byte order: big endian
Speed over ground	40213	00D5	Float (4Byte)	Unit is knots, read-only, word order: big endian, byte order: big endian
Course over ground	40215	00D7	Float (4Byte)	Units in degrees, read-only, word order: big endian, byte order: big endian
Antenna Settings	40217	00D9	Int16	0: active antenna; 1: passive antenna
Satellite system selection	40218	00DA	Int16	The satellite systems supported by different models vary. For details, please refer to the description below
Location update frequency	40219	00DB	Int16	0x0000: 1Hz, 0x0001: 2Hz, 0x0002: 5Hz, 0x0003: 10Hz Note: The EWD108-GN05 series supports: 0x0000: 1Hz, 0x0001: 10Hz

IAP Upgrade Logo	40220	00DC	Int16	Write to 0x01, enter IAP upgrade mode, does not support readout, using standard YMODEM protocol, it is recommended to use the IAP firmware upgrade function with the host computer
Product model	40221	00DD	String (14Byte)	ASCII encoding is used and AB is used for the decoding order
Version information	40228	00E4	String (14Byte)	ASCII encoding is used and AB is used for the decoding order

Note: Single-precision floating point uses the standard IEEE754 format with 32 bits (4 bytes). The default single-precision floating-point size-end mode is ABCD (high byte first, low byte second), e.g. 0x3FF1EB85 represents 1.89 (2 decimal places reserved).

Model	Register setting values correspond to the satellite system
EWD108-GN03(xxx)	0x0000 : GPS 0x0001 : BDS 0x0002 : GPS+BDS(default) 0x0003 : GLONASS 0x0004 : GPS+GLONASS 0x0005 : BDS+GLONASS 0x0006 : GPS+BDS+GLONASS
EWD108-GN04(xxx)	0x0000 : GPS L1CA \SBAS L1CA \QZSS L1CA L1S 0x0001 : GPS L1CA \SBAS L1CA \BeiDou B1 \QZSS L1CA L1S 0x0002 : GPS L1CA \SBAS L1CA \Galileo E1 \BeiDou B1 \QZSS L1CA L1S(default) 0x0003 : GPS L1CA \SBAS L1CA \QZSS L1CA L1S \GLONASS L1 0x0004 : GPS L1CA \SBAS L1CA \Galileo E1 \QZSS L1CA L1S 0x0005 : GPS L1CA \BeiDou B1 0x0006 : GPS L1CA \SBAS L1CA \Galileo E1 \QZSS L1CA L1S \GLONASS L1
EWD108-GN05(xxx)	0x0000 : GPS L1+L5 0x0001 : BDS B1I+B2A 0x0002 : GPS L1+L5 \BDS-B1I+B2A 0x0003 : GPS L1+L5 \GLONASS G1 0x0004 : GPS L1+L5 \QZSS L1+L5 0x0005 : GPS L1+L5 \GALIEO E1+E5A 0x0006 : GPS L1+L5 \BDS B1I+B2A \GLONASS G1 \QZSS L1+L5 \GALIEO E1+E5A \SABA L1 (default)
EWD108-GN03B(xxx)	Only BDS is supported, and this setting cannot be changed.
EWD108-GN06B(xxx)	Only BDS is supported, and this setting cannot be changed.

4 Quick Start

4.1 Preparation work

Note: EWD108-GN0x, stands for EWD108-GN03/04/05/03B/06B five models collectively.

Take the example of obtaining the RMC positioning information output by the device.

In order to test the EWD108-GN0x positioning module, the following hardware is required:

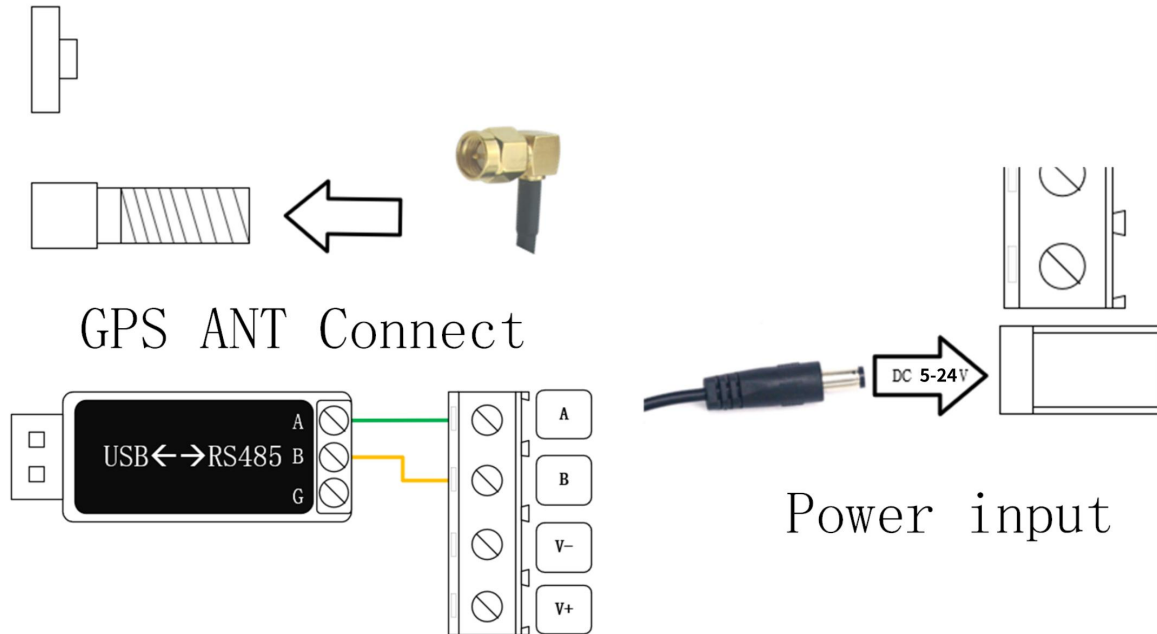
- One PC;
- EWD108-GN0x (485) positioning device. For the demonstration, an RS485 interface device is used here. For other interfaces, please refer to the actual situation;
- Active antenna or passive antenna. The following is explained with the active antenna (SMA, internal thread, internal pin);
- One USB to RS485 serial cable;

		
PC	EWD108-GN0x(485)	Threads
		
antenna	USB to RS485	DC 12V Power supply

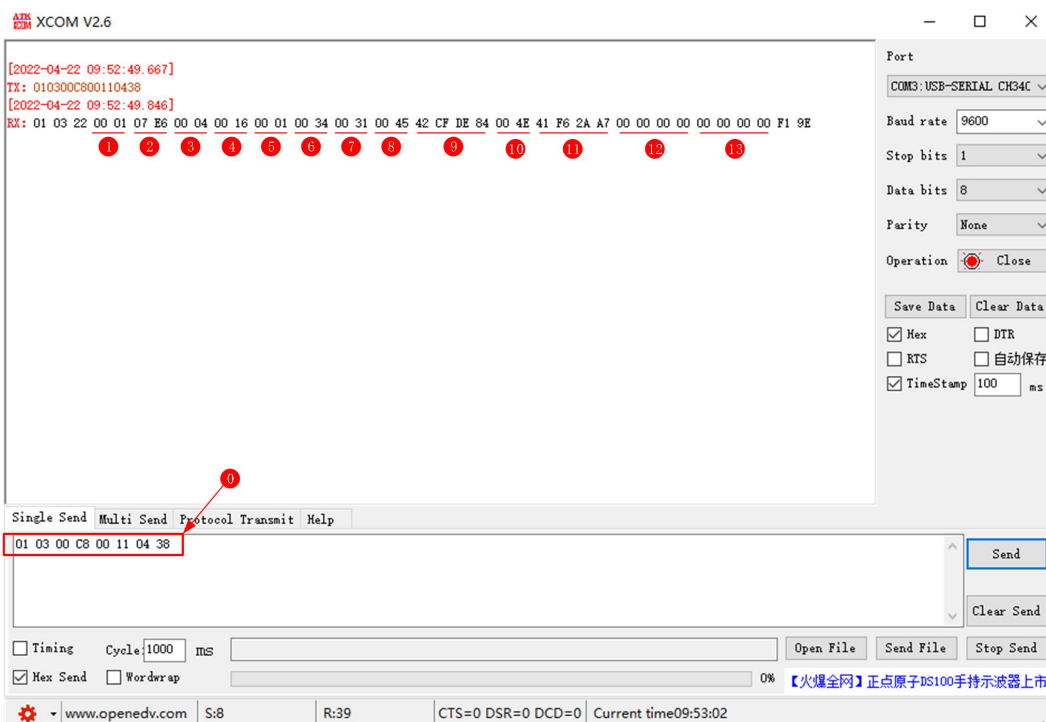
To prepare the serial debugging tool, you can get the 'XCOM' serial debugging tool from the 'Related downloads' of the corresponding products in the official website of EBYTE (www.cdebyte.com);

4.2 Demonstration of use

After preparing the above hardware and software, you can query the positioning information of the device and the UTC time through Modbus RTU command, connect the power supply of the device and the serial port correctly, connect the antenna and move the antenna to an open place, as shown in the figure below:



Select the correct serial port number and adjust the serial port parameters to 9600bps, 8N1. Send the 16-bit data "01 03 00 C8 00 11 04 38" to read the positioning data. You can receive the device response data as shown in the figure below. If you cannot receive the response data, check the power supply or ensure the correct connection of the serial port. Also, check if the parameter configuration of the serial port debugging assistant is correct. If the configuration is correct, you can use the "factory reset button" to restore the factory parameters and then test again.



Data Parsing Table

Serial number	Original value (HEX)	Descriptions	Conversion rules	after conversion
---------------	----------------------	--------------	------------------	------------------

1	0x00 00	Position effectiveness	0x00: Invalid 0x01:Valid	Valid
2	0x07 e6	Year	HEX→DEC	Year 2022
3	0x00 04	Month	HEX→DEC	April
4	0x00 16	Day	HEX→DEC	22nd
5	0x00 01	Hour	HEX→DEC	1
6	0x00 34	Minute	HEX→DEC	34
7	0x00 31	Seconds	HEX→DEC	31
8	0x00 45	Direction of Longitude	Low effective HEX→ASCII	E(longitude east)
9	0x42 cf de 84	Longitude	32-bit Floating Point, Big End - Big End	103.93460083007812
10	0x00 4e	Direction of Latitude	Low effective, HEX→ASCII	N(latitude north)
11	0x4e f6 2a a7	Latitude	32-bit Floating Point, Big End - Big End	30.77082633972168
12	0x00 00 00 00	Speed over ground	32-bit Floating Point, Big End - Big End	0
13	0x00 00 00 00	Course over ground	32-bit Floating Point, Big End - Big End	0

[Note]: The time is UTC time, and 8 hours need to be added to convert to Beijing time.

5 Modbus RTU Usage

Note: The following demonstration instructions have a device address of 1. If other addresses are used, the address bits are different from the checksum.

5.1 Holding registers

The registers used by the EWD108-GN0x device are all holding registers, and the function codes for writing holding registers are 0x06 (write a single holding register), and 0x03 (read a holding register) is used for reading holding registers. 0x03 code read instruction format (read baud rate for example):

Device address	Function code	First address	Number of readings	Check digit CRC
01	03	00 03	00 01	74 0A

Return format (to read baud rate for example):

Device address	Function code	Data length	Number of readings	Check digit CRC
01	03	02	00 03	F8 45

0x06 Code Configuration Command Format (configure baud rate as an example):

Device address	Function code	First address	Value	Check digit CRC
01	06	00 03	00 02	F8 0B

5.2 Read Holding Register Instruction

5.2.1 Read reserved bit 0001

Command Frame: 01 03 00 01 00 01 D5 CA

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0x01	0x00 0x01	0xD5 0xCA

Response frame: 01 03 02 00 10 B9 88

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x56	0x38 0x7A

Description:

The 0x56 in this return data has no practical significance and is for read only.

5.2.2 Read device address (broadcast)

[Note] This command is a general-purpose read command for addresses, using broadcast commands. To avoid conflicts with other devices in the system, ensure that only the device to be read is connected to the bus when reading.

Command Frame: 00 03 00 02 00 01 24 1B

Address	Function code	Register Starting Address	Number of registers	CRC check
0x00	0x03	0x00 0x02	0x00 0x01	0x24 0x1B

Response frame: 00 03 02 00 01 44 44

Address	Function code	Data length	Data	CRC check
0x00	0x03	0x02	0x00 0x01	0x44 0x44

Description:

This command is a general-purpose read command for the address, using the broadcast command. In order to avoid conflicts with other devices in the system, the read ensures that only the device to be read is connected to the bus.

5.2.3 Reading device baud rate

Command Frame: 01 03 00 03 00 01 74 0A

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0x03	0x00 0x01	0x74 0x0A

Response frame: 01 03 02 00 03 F8 45

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x03	0xF8 0x45

Description:

Return baud rate is 0x03, which represents 9600 bps, the rest of the baud rate codes are described in ‘Modbus Register Table’.

5.2.4 Parity Reading

Command Frame: 01 03 00 04 00 01 C5 C8

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0x04	0x00 0x01	0xC5 0xC8

Response frame: 01 03 02 00 00 B8 44

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x00	0xB8 0x44

Explanation: The return parity bit is 0x00, which means no parity, and the rest of the parity code is described in ‘Modbus Register Table’.

5.2.5 Reading positioning data (RMC)

Command frame: 01 03 00 05 00 23 14 12

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0x05	0x00 0x23	0x14 0x12

Response Frame:

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x46	70 bytes of data	2-byte checksum

Positioning data (RMC) parsing

The 70 bytes of data returned by the Read Locate Data (RMC) conforms to the NMEA0183 protocol and is displayed in ASCII as follows:

\$GNRMC,083429.00,a,3046.26769,n,10356.04948,e,000.00,089.80,190422*21

Field	Symbol	Meaning	Range of values	Example	Remark
1	\$				
2	GNRMC				RMC protocol header, GNRMC for joint positioning
3	hhmmss.ss	UTC time	hour minute second.second	072905.00	Beijing East 8 needs +8
4	A	position state	A/V		A-valid, V-ineffective
5	ddmm.mmmmm	longitude	degree degree cent cent. Cent cent cent cent	3640.46260	Calculation to be converted to degrees: 36 degrees +

					40.46260 cents. 40.46260/60 = 0.67438 degrees, so 36.67438 degrees
6	a	latitudinal direction	N/S		N-North, S-South
7	ddmm.mmmmm	longitudes	degree degree cent cent. Cent cent cent cent	3640.46260	Calculation to be converted to degrees: 36 degrees + 40.46260 cents. 40.46260/60 = 0.67438 degrees, so 36.67438 degrees
8	a	longitude direction	E/W		E - Eastern Longitude, W - Western Longitude
9	xxx.xx-xxx.xx	speed over ground	knot	123.2	Ground rate in knots, range 000.00 to 999.99 knots, zero if leading digit is insufficient
10	xxx.xx-xxx.xx	course over earth	degree	000.0~359.9	Ground heading (000.00 to 359.99 degrees, with true north as the reference point), if the leading digit is insufficient, it is zero.
11	xxxxxx	Date	day, month and year	190422	19 April 2022
13	*	statement terminator			
14	24	checksum	The data between '\$' and '*' (excluding these two characters) is performed on a byte-by-byte basis as a different-or operation, expressed as a hexadecimal number		

5.2.6 Reading antenna settings

Command frame: 01 03 00 D9 00 01 55 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0xD9	0x00 0x01	0x55 0xF1

Response frame: 01 03 02 00 00 B8 44

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x00	0xB8 0x44

Explanation: The return parity bit is 0x00, which means active antenna is used, the rest is described in ‘Modbus register table’.

5.2.7 Reading the product model

Command Frame: 01 03 00 E1 00 06 95 FE

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0xDD	0x00 0x07	0x94 0x32

Response frame: 01 03 0E 45 57 44 31 30 38 2D 47 4E 30 35 20 0D 0A 69 4B

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x0E	14bytes of data	0x69 0x4B

Explanation: Take EWD108-GN05 for example, return 14 bytes of valid data as ‘EWD108-GN05\r\n’.

5.2.8 Reading version information

Command Frame: 01 03 00 E7 00 07 B4 3F

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0xE4	0x00 0x07	0x44 0x3F

Response frame: 01 03 0E 46 57 2D 37 35 30 32 2D 32 2D 31 32 0D 0A 52 0D

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x0E	14 bytes of data	0x52 0x0D

Note: The 14-byte data returned is ‘FW-7XXX-X-XX\r\n’, which is different for different models, and is only for reference.

5.2.9 Reading the Satellite System

Command Frame: 01 03 00 DA 00 01 A5 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0xDA	0x00 0x01	0xA5 0xF1

Response frame: 01 03 02 00 06 38 46

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x06	0x52 0x0D

Note:Based on the EWD108-GN05 series reference, the return value is 0x06, indicating that the satellite system used is Mode 6. For more details, please refer to the description of the aforementioned satellite system.

5.2.10 Read the update frequency of GNSS positioning

Command Frame: 01 03 00 DB 00 01 F4 31

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x03	0x00 0xDB	0x00 0x01	0xF4 0x31

Response frame: 01 03 02 00 00 B8 44

Address	Function code	Data length	Data	CRC check
0x01	0x03	0x02	0x00 0x00	0x52 0x0D

Note: The return positioning update frequency is 0x00, indicating that the RTU's positioning update frequency is 1Hz.

5.3 Write Holding Register Instructions

5.3.1 Modify device address (broadcast)

[Note] This command is a general-purpose write command for addresses, using broadcast commands. To avoid conflicts with other devices in the system, ensure that only the device to be configured is connected to the bus when reading.

Command Frame: 00 06 00 02 00 01 E8 1B

Address	Function code	Register Starting Address	Number of registers	CRC check
0x00	0x06	0x00 0x02	0x00 0x01	0xE8 0x1B

Response frame: 00 06 00 02 00 01 E8 1B

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0x02	0x00 0x01	0xE8 0x1B

Description:

This command is used to set the device address, use 0x00 as the broadcast address and change the device address to 0x01.

5.3.2 Modifying the baud rate

Command Frame: 01 06 00 03 00 03 39 CB

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0x03	0x00 0x03	0x39 0xCB

Response frame: 01 06 00 03 00 03 39 CB

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0x03	0x00 0x03	0x39 0xCB

Description:

This command sets the baud rate of the device to 9600.

The default baud rate of the device is 9600 without parity, users can set the baud rate and parity according to the actual demand.

5.3.3 Modifying the parity bit

Command Frame: 01 06 00 06 00 04 09 CB

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0x06	0x00 0x04	0x09 0xCB

Response frame: 01 06 00 06 00 04 09 CB

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0x06	0x00 0x04	0x09 0xCB

Description:

This command sets the parity bit of the device to odd parity.

5.3.4 Modifying antenna settings

Command Frame: 01 06 00 D9 00 01 99 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xD9	0x00 0x01	0x99 0xF1

Response frame: 01 06 00 D9 00 01 99 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xD9	0x00 0x01	0x99 0xF1

Description:

This command is used to set the antenna as passive antenna.

5.3.5 Modifying the IAP flag

Command Frame: 01 06 00 DC 00 01 89 F0

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xDC	0x00 0x01	0x89 0xF0

Description:

This command is used to write the IAP upgrade flag, after writing, it will immediately reset into the IAP procedure.

5.3.6 Modifying the Satellite System

Command Frame: 01 06 00 DA 00 06 28 33

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xDA	0x00 0x06	0x28 0x33

Response frame: 01 06 00 DA 00 06 28 33

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xDA	0x00 0x06	0x28 0x33

Description:

After this command is written, enable Satellite System Mode 6. Please refer to the above instructions for details.

5.3.7 Adjust the update frequency of the GNSS module

Command Frame: 01 06 00 DB 00 00 F9 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xDB	0x00 0x00	0xF9 0xF1

Response frame: 01 06 00 DB 00 00 F9 F1

Address	Function code	Register Starting Address	Number of registers	CRC check
0x01	0x06	0x00 0xDB	0x00 0x00	0xF9 0xF1

Description:

This command is used to set the update frequency of the GNSS module to 1Hz.

Revision history

Version	Date	Description	Issued by
1.0	2024-12-10	Initial version	Bin
1.1	2025-08-14	Modify pin and indicator light descriptions	Bin
1.2	2025-08-27	Content revision	Bin
1.3	2025-10-13	Newly added EWD108-GN03B(xxx) and EWD108-GN06B(xxx) series products; Newly added some instructions; Optimize the format of the manual.	Bin

About us

Hot line:4000-330-990

Technical support: service@cdebyte.com

Documents and RF Setting download link: <https://www.cdebyte.com>

Thank you for using Ebyte products! Please contact us with any questions or suggestions: info@cdebyte.com

Address: B2 Mould Park, 199# Xiqu Ave, High-tech District, Sichuan, China

